Q.1. Availability of energy is less for entities at higher trophic levels. Why?

A.1. The 10% energy flow law suggested by Lidman, is followed in the ecosystem. As per this law, only 10% of the energy that is available at every trophic level is transferred to the next trophic level. The remaining is lost in the form of heat (during respiration). As we approach the higher levels, energy keeps declining that is available to entities. Hence the topmost carnivore attains the least amount of energy in a food chain.

Q.3. Can we address an aquarium as a complete ecosystem?

A.3. An aquarium is an artificial ecosystem made by man. An ecosystem is said to be complete if it possesses all the biological and physical components vital for the survival of fishes. Hence it is a complete ecosystem.

Q.6. Explain why the flow of energy at different levels in an ecosystem is unidirectional and non-cyclic.

A.6. The flow of energy is one way and non-cyclic(Plants→Herbivores→Carnivores→Top Carnivore). The energy that is transferred from 1st to the next trophic level decreases with each passing level and cannot flow in the reverse direction.

Q.4. Explain how energy flow supports the second law of thermodynamics in an ecosystem.

A.4. As per the second law of thermodynamics, every activity that includes transformation includes wastage of energy in the form of heat and rise in disorganization excluding deep hydrothermal ecosystems. Out of the total PAR(Photosynthetically Active Radiation), only 2-10% is absorbed by organisms involved in photosynthesis to produce organic matter. Furthermore, the energy is utilized in metabolic activities, to form food and in storing biomass(very less). Biomass or the trapped energy is passed to the next trophic level as per the Lindeman's law. 10% of the stored energy is transferred from one to the next consecutive trophic level.

Q.5. Answer the following questions. Write the outcomes of the following events.

a) The consequence of eliminating all producers

b) The consequence of eliminating all entities at the herbivore level

c) The consequence of eliminating all top carnivore entities

A.5. a) It diminishes primary production in an ecosystem and hence unavailability of biomass to higher trophic levels b) It would result in an increase in primary productivity and biomass of producers. Carnivorous animals, due to unavailability of food, will not survive. c) There will be an increase in the herbivore population, resulting in over-grazing and hence desertification.

Question 66.

Rajni was watching a programme on TV about the earth. The host was explaining how the increased cutting of trees and forests is affecting the ecosystem, its components and services, etc. She got curious and asked her teacher about 'what services the ecosystem provides to human beings?

(i) How does ecosystem services benefit us?

(ii) Name the value put forth by Robert Constanza and his colleagues on ecosystem services.

(iii) What values are showed by Rajni?

Answer:

(i) Healthy ecosystems provide wide range of economic, environmental, aesthetic goods and services. These services include purification of air, mitigation of droughts and floods, cycling of nutrients, providing habitats for wildlife, maintenance of biodiversity, etc.

(ii) Robert Constanza and his colleagues have estimated the average value of fundamental ecosystem service at US $ 33 trillion a year.

(iii) Values shown by Rajni are awareness and curiosity towards environmental affairs.

Question 65.

Aryan worked on the farm with his father.

He saw that his father added earthworms to the soil. He came to his father and told him not to do so. His father explained why he was doing so and Aryan was satisfied with the explanation.

(i) Why were earthworms added to the soil?

(ii) What values do you observe in Aryan's father?

Answer:

(i) Earthworms are detritivores which cause the breakdown of detritus into small, simpler forms. This helps in improving soil quality.

(ii) Aryan's father is intelligent, hard worker and environment friendly.

Question 64.

In a food chain, grass is consumed by goat which in turn is consumed by humans. Anant states that man is the secondary consumer in this food chain. Is Anant correct? It yes, give reason.

Answer:

Yes, Anant is correct. In this food chain grass is the producer which feeds energy into the food chain and goat is the primary consumer. Since, goat is eaten by humans, they are called secondary consumers.

Question 56.

Discuss the role of healthy ecosystem services as a prerequisite for a wide range of economic, environmental and aesthetic goods and services. (Delhi 2017)

Answer:

Ecosystem services are the products of ecosystem processes. Forests are the major source of ecosystem services and are prerequisite for environmental, aesthetic goods and indirect economic values in the following ways

1. Environmental Values:

Carbon-fixation Huge amount of $CO2$ in the atmosphere is removed naturally and fixed by plants intoorganic molecules and energy through photosynthesis. All the other trophic levels, i.e. consumers depend upon this energy produced by them.

Release of oxygen by the producers as a byproduct in the process of photosynthesis, improves the air quality and supports life on earth.

Soil formation and soil protection are the major ecosystem services accounting for nearly 50% of their total worth. Plant cover protects the soil from drastic changes in the temperature. There is little wind or water erosion as soil particles are not exposed to them. The soil remains spongy and fertile. There occurs no landslides and floods.

Nutrient cycling There is no depletion of nutrients, but the same are repeatedly circulated.

Question 44.

(i) Explain the significance of ecological pyramids with the help of an example.

(ii) Why are the pyramids referred to as upright or inverted? Explain. (All India 2012)

Answer:

(i) Significance of ecological pyramids: They graphically represent the relation between producers and consumers in order to calculate energy content, biomass and number of organisms of that trophic level. A

trophic level represents only a functional level not a species as such. A given species may occupy more than one trophic level in the

same ecosystem at the same time. The ecological pyramids provide an overall idea of the trophic levels occupied by an organism in an ecosystem.

Example A sparrow is a primary consumer when it eats seeds, fruits, peas and a secondary consumer when it eats insects and worms.

(ii) Upright pyramids: When producers are more in number and biomass than the herbivores and herbivores are more in number and biomass than the carnivores, pyramids are referred to as upright.

Energy at a lower trophic level is always more than at a higher trophic level. Pyramid of energy is referred to as always upright.

Inverted pyramids When the numbers and biomass of producers are less and consumers increase and become largest in top consumer level, Pyramid of number and biomass is referred to as inverted.

Question 42.

(i) With suitable examples, explain the energy flow through different trophic levels. What does each bar in this pyramid represent?

(ii) Write any two limitations of ecological pyramids. (Delhi 2014C)

Answer:

(i) The energy flows unidirectionally from the first trophic level (producers) to last trophic level (consumers) and as the energy flows from one trophic level to another, some energy is always lost as heat into the surrounding environment. So, the amount of energy flowing decreases at each successive trophic level. This can be explained with the help of a diagram of a grazing food chain. Refer to text on page no. 352.

The pyramid of energy is always upright and each bar in the pyramid indicates the amount of energy present at each trophic level in a given time or per unit area.

(ii) The limitations of ecological pyramids are

It does not consider the same single species operating at two or more trophic levels.

It assumes simple food chains that do not exist in nature and do not accommodate food web.

Saprophytes, detritivores and decomposers are not given any place in pyramids, despite their vital role in ecosystem (any two).

Bio-diversity

Que: How is biodiversity measured?
Ans: Biodiversity is determined by counting the number of species occurring in a given unit of area. The greater the species diversity within an area, the higher the biodiversity, which can be calculated using various methods, such as diversity indices.

Que. Why is biodiversity important to the planet?
Ans. Biodiversity is important to the planet, because healthy, diverse ecosystems are better equipped to withstand and recover from a variety of disasters. All the species in an area work together to survive and maintain their ecosystem. For instance, large carnivores, such as lions, leopards, and wolves are essential for a healthy ecosystem. If the population of such carnivores falls, the herbivores they prey upon will likely see a population increase. This results in environmental deterioration, because they graze more. Thus a healthy ecosystem depends on a delicate balance based on cooperation and mutual survival.

Que. What are the main causes of biodiversity loss?
Ans. As you've probably guessed by now, human activities threaten biodiversity. There are five main causes of biodiversity loss:
Changes in land and sea use
Overexploitation
Climate change
Pollution
Invasive alien species
Worldwide, more than 33% of the Earth's land surface and nearly 75% of freshwater resources are now devoted to crop or livestock production. In the Amazon Rainforest, the WWF finds that 80% of current deforestation is due to extensive cattle ranching, while agriculture is largely responsible for the rest.

Que. Differentiate between species diversity and ecological diversity.
Ans. The distribution and number of species in an area are referred to as species diversity whereas ecological diversity describes the diversity at the ecosystem level.

Que. State two ways through which humans are benefitted from biodiversity.
Ans. We derive economic benefits from the diversity of entities, such as:
Food, fibre, firewood
Industrial products, construction material
Medicinal products from plants
Pure oxygen, flood and soil erosion control, natural pollinators
Recycling of wastes by microbes
Nutrient restoration

Que. How can the loss of biodiversity be prevented?

Ans. The occurrence of different types of habitat, species, ecosystem, gene pool, and a gene in a particular area in biodiversity. It can be conserved with various conservational strategies and management of abiotic and biotic resources. Listed below are a few conservational strategies:

Natural conservation or protection of useful plants and animals in their natural habitats.

Conserving crucial habitats like breeding and feeding areas, facilitating the growth and multiplication of endangered species

Regulation or banning hunting activities

Through bilateral or multilateral agreements, habitats of migratory entities should be conserved

Spreading awareness of the significance of conservation of biodiversity

Avoiding over-exploitation of natural resources